Haithem Turki

Curriculum Vitae

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Education

Carnegie Mellon University **2019 - 2024**
Ph.D. in Computer Science
Thesis: Towards City-Scale Neural Rendering
Advisor: Deva Ramanan (committee chair)
Committee: Jessica K. Hodgins, Martial Hebert, Jonathan T. Barron

Stanford University **2012 - 2018**
M.S. in Computer Science

Stanford University **2008 - 2012**
B.S. in Computer Science, minor in Management Science and Engineering

Experience

NVIDIA Research, Senior Research Scientist **2024-**
AI-Mediated Reality and Interaction (AMRI) group with Shalini De Mello

Meta, Research Scientist Intern w/ Christian Richardt, Michael Zollhöfer **2023**
Argo AI, Research Intern w/ Deva Ramanan, Francesco Ferroni **2022**
Page One Ventures, Venture Partner **2020 - 2024**
Palantir Technologies, Engineering Manager and Technical Lead **2012 - 2019**
Addepar, Engineering Intern **2011**
Stanford University, Research Assistant w/ Daphne Koller, M. Pawan Kumar **2010**

Awards & Honors

Oral Paper, NeurIPS 2026 (top 0.4% of submissions) **2026**
Déjà View: Looping Transformers for Multi-View 3D Reconstruction

Best Paper Award Candidate, CVPR 2025 **2025**
DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models

Highlight Paper, CVPR 2024 (top 2.8% of submissions) **2024**
HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces

Highlight Paper, CVPR 2024 (top 2.8% of submissions) **2024**
SpecNeRF: Gaussian Directional Encoding for Specular Reflections

Publications

[18] Alessandro Burzio*, Tobias Fischer*, Sven Elflein, Qunjie Zhou, Riccardo de Lutio, Jiawei Ren, Jiahui Huang, Shengyu Huang, Marc Pollefeys, Laura Leal-Taixé, Zan Gojcic†, **Haithem Turki†**,“Déjà View: Looping Transformers for Multi-View

3D Reconstruction" in Conference on Neural Information Processing Systems (NeurIPS), 2026, ***(Oral)***

[17] Riccardo de Lutio*, Tobias Fischer, Yen-Yu Chang, Yuxuan Zhang, Jay Zhangjie Wu, Xuanchi Ren, Tianchang Shen, Katarina Tothova, Zan Gojcic, **Haithem Turki***,"ArtiFixer: Enhancing and Extending 3D Reconstruction with Auto-Regressive Diffusion Models" in ACM SIGGRAPH, 2026, ***(Invited Oral, GenRecon3D Workshop @ CVPR 2026)***

[16] Yuxuan Zhang*, Katarina Tothova*, Zian Wang, Kangxue Yin, **Haithem Turki**, Riccardo de Lutio, Yen-Yu Chang, Or Litany, Sanja Fidler, Zan Gojcic,"DiffusionHarmonizer: Bridging Neural Reconstruction and Photorealistic Simulation with Online Diffusion Enhancer" in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026

[15] **Haithem Turki***, Qi Wu*, Xin Kang, Janick Martinez Esturo, Shengyu Huang, Ruilong Li, Zan Gojcic, Riccardo de Lutio,"SimULi: Real-Time LiDAR and Camera Simulation with Unscented Transforms" in International Conference on Learning Representations (ICLR), 2026

[14] Sherwin Bahmani, Tianchang Shen, Jiawei Ren, Jiahui Huang, Yifeng Jiang, **Haithem Turki**, Andrea Tagliasacchi, David B. Lindell, Zan Gojcic, Sanja Fidler, Huan Ling, Jun Gao*, Xuanchi Ren*,"Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation" in International Conference on Learning Representations (ICLR), 2026

[13] Zhangjie Wu*, Yuxuan Zhang*, **Haithem Turki**, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic†, Huan Ling†,"DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models" in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025, ***(Best Paper Award Candidate)***

[12] **Haithem Turki**, Vasu Agrawal, Samuel Rota Bulò, Lorenzo Porzi, Peter Kontschieder, Deva Ramanan, Michael Zollhöfer, Christian Richardt,"HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces" in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, ***(Highlight)***

[11] Li Ma, Vasu Agrawal, **Haithem Turki**, Changil Kim, Chen Gao, Pedro Sander, Michael Zollhöfer, Christian Richardt,"SpecNeRF: Gaussian Directional Encoding for Specular Reflections" in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, ***(Highlight)***

[10] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan,"PyNeRF: Pyramidal Neural Radiance Fields" in Conference on Neural Information Processing Systems (NeurIPS), 2023

[9] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan,"Low-Bandwidth Self-Improving Transmission of Rare Training Data" in Annual International Conference on Mobile Computing And Networking (MobiCom), 2023

[8] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan,"Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data" in International Workshop on Networked AI Systems (NetAISys), 2023

[7] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan,"SUDS: Scalable Urban Dynamic Scenes" in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023

[6] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai,"Accelerating Silent Witness Storage" in IEEE Micro, Nov.-Dec. 2022

[5] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan,“Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022

[4] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan,“OpenRTiST: End-to-End Benchmarking for Edge Computing” in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, Oct.-Dec. 2020

[3] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng,“Edge Computing for Legacy Applications” in IEEE Pervasive Computing, Oct.-Dec. 2020

[2] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller,“Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation” in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, July 2015

[1] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller,“Learning Specific-Class Segmentation from Diverse Data” in International Conference on Computer Vision (ICCV), 2011

Technical Reports

[3] Aarti Basant, Amlan Kar, Despoina Paschalidou, Fangyin Wei, Francesco Ferroni, Guillermo Garcia Cobo, **Haithem Turki**, Huan Ling, Jaewoo Seo, James Lucas, Jay Zhangjie Wu, Jialiang Wang, Jonathan Lorraine, Jun Gao, Kai He, Katarina Tothova, Kevin Xie, Michał Tyszkiewicz, Qi Wu, Riccardo de Lutio, Ruilong Li, Sanja Fidler, Seung Wook Kim, Tianchang Shen, Tianshi Cao, Tobias Pfaff, William Lew, Xindi Wu, Xuanchi Ren, Yifan Lu, Yuxuan Zhang, Zan Gojcic, Zian Wang,“NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation”, arXiv:2606.03159, 2026

[2] Tianshi Cao*, Jiawei Ren*, Yuxuan Zhang, Jaewoo Seo, Jiahui Huang, Shikhar Solanki, Haotian Zhang, Mingfei Guo, **Haithem Turki**, Muxingzi Li, Yue Zhu, Sipeng Zhang, Zan Gojcic, Sanja Fidler, Kangxue Yin,“Asset Harvester: Extracting 3D Assets from Autonomous Driving Logs for Simulation”, arXiv:2604.18468, 2026

[1] Ziqiang Feng, Shilpa George, **Haithem Turki**, Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan,“Improving Edge Elasticity via Decode Offload”, Technical Report CMU-CS-21-139, Carnegie Mellon University, 2021

Patents

[2] **Haithem Turki**, Robert Fink, Amr Al Mallah,“Systems and Methods for Indexing and Searching.” US Patent 2019/0347343 A1, 2019

[1] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara,“Systems and Methods for Constraint Driven Database Searching.” US Patent 2018/293239 A1, 2018

Teaching

15-640 Distributed Systems, Carnegie Mellon University (Graduate Student Instructor) 2020 - 2021

17-700 Data Science and Machine Learning at Scale, Carnegie 2020 - 2021

Mellon University (Graduate Student Instructor)
15-821 Mobile and Pervasive Computing, Carnegie Mellon University (Project Mentor) 2020 - 2021

Student Collaborators & Interns

Tobias Fischer, PhD Student @ ETH Zürich. See Publications [17, 18].
Alessandro Burzio, PhD Student @ University of Modena and Reggio Emilia. See Publications [18].
Yen-Yu Chang, PhD Student @ Cornell University. See Publications [16, 17].

Invited Talks

"Photoreal Worlds at Scale: From Capture to Generation"
EPFL 2026
ETH Zürich 2026
ZurichCV (ZurichAI) 2026
Caltech 2026

"Towards City-Scale Neural Rendering"
Stanford Vision Lab 2024
Waymo Research (virtual) 2024
Google (virtual) 2024
NVIDIA Toronto AI Lab (virtual) 2024

"Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs"
Argo AI 2022

Academic Service

Area Chair:
ACCV

Reviewer:
CVPR, ACCV, ECCV, ICCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, AAAI, Transactions on Image Processing

Additional Information

Languages:
English (native), French (fluent), Arabic (fluent), Japanese (elementary), German (elementary)

Other Qualifications:
Alchemist Accelerator (startup mentor and investor). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.